

India Coffee Market Intelligence — ML Projects

Engineering student guide • Clustering, Classification, Regression, Forecasting & Recommendation

Project principle: ML estimates market opportunity and consumer behaviour; it cannot guarantee commercial success.

1. Business Problem

A tea-producing company in India plans to enter coffee with a new brand and a partnership with Brazilian coffee company 3 Corações. Before launch, the company wants data-driven answers about consumer segments, likely buyers, spending, market trends, cities and competitive pressure.

2. ML Project Architecture

Consumer/market data → EDA → Cleaning → Feature engineering → ML models → Validation → Segments + predictions → Opportunity analysis → Launch recommendation

3. Consumer Segmentation

Type: Unsupervised Learning | **Algorithms:** K-Means / Agglomerative Hierarchical Clustering

Useful variables: age, city, income, coffee frequency, tea frequency, monthly coffee spend, coffee format, café visits, price sensitivity and premium preference.

Possible data-driven segments

Convenience-oriented • Café/experience consumers • Specialty-coffee enthusiasts • Value-sensitive consumers. Students must name clusters only after examining their statistics.

Student outputs

Cluster visualizations • dendrogram (hierarchical) • silhouette score • cluster profiles • business interpretation

4. New Coffee Brand Adoption Prediction

Type: Classification | **Target:** Will_Buy_New_Brand (0/1)

Predict the probability that a consumer will purchase the proposed brand. The target should come from a properly designed survey or historical purchase/campaign data.

Candidate features

Age • Income • Coffee Frequency • Coffee Spend • Current Brand • Café Visits • Price Sensitivity • Premium Preference • Origin Interest • Product Format

Models

Logistic Regression • Decision Tree • Random Forest • Gradient Boosting/XGBoost

Evaluation

Accuracy • Precision • Recall • F1 • ROC-AUC • Confusion Matrix • Probability calibration

5. Coffee Spending Prediction

Type: Regression | **Target:** Monthly Coffee Spend

Estimate expected monthly expenditure using demographic and behavioural variables. Metrics: MAE, RMSE and R².

6. Coffee Market / Demand Trend Forecasting

Type: Time Series / Regression

Use time-indexed sales, search interest, survey demand or other lawful market indicators to study trends in coffee, specialty coffee, cold coffee or RTD coffee.

Methods

Moving Average • Linear Trend • ARIMA • Prophet • LSTM (advanced)

7. City-wise Coffee Market Opportunity

Type: Clustering / Scoring / Classification

Construct city-level features such as coffee interest, young population share, income indicators, café density, premium spending, e-commerce penetration and competitor presence.

Example output

Cluster A → Premium coffee markets Cluster B → Emerging coffee markets Cluster C → Mass/instant-oriented markets

8. Market Entry Recommendation Engine

Combine evidence from segmentation, adoption probability, spending potential, trends and city opportunity into a transparent ranking model. Weights must be documented and validated rather than invented.

9. Testing the Brazilian Partnership

Treat the 3 Corações partnership as a hypothesis. A controlled survey experiment can compare purchase intention with and without partnership/origin information.

Hypothesis: Partnership information increases purchase intention among selected consumer groups.

10. Recommended Final-Year Project

AI-Based Coffee Market Intelligence and Consumer Adoption Prediction for a New Coffee Brand in India

Modules: Consumer Segmentation → Adoption Classification → Spending Regression → Trend Forecasting → City Opportunity → Market Entry Recommendation

11. Final Deliverables

Notebook(s) • cleaned data • EDA • trained models • evaluation • cluster profiles • predictions • city/segment ranking • README • report • presentation • optional Streamlit dashboard

12. Responsible Framing

Do not claim guaranteed market share or success without validated market-sizing evidence. Present predictions as estimates and clearly state limitations.